Matrices (AI HL)

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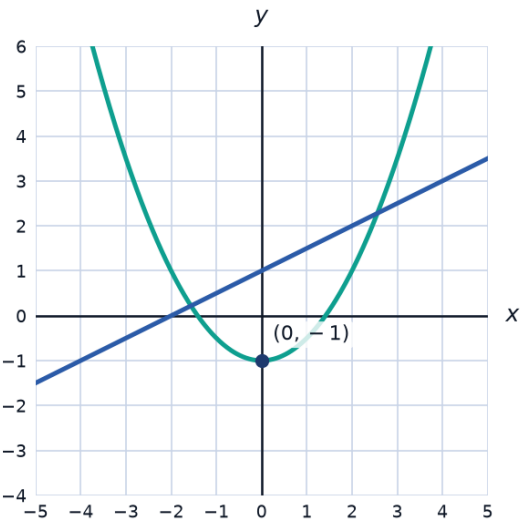
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Matrix Multiplication and Inverses**

A transition matrix is $T = \left(\frac{4}{5}, \frac{3}{5}; \frac{1}{5}, \frac{2}{5}\right)$, rows/cols ordered A, B. Find $P(B|B)$, the probability of moving from state B to state B.

2 **[Foundational]** • **Target: Determinants and Singularity**

A transition matrix is $T = \left(\frac{4}{5}, \frac{2}{5}; \frac{1}{5}, \frac{3}{5}\right)$, rows/cols ordered A, B. Find $P(A|A)$, the probability of moving from state A to state A.



3 **[Foundational]** • **Target: Eigenvalues and Eigenvectors**

A transition matrix is $T = \left(\frac{2}{5}, \frac{3}{5}; \frac{3}{5}, \frac{2}{5}\right)$, rows/cols ordered A, B. Find $P(B|A)$, the probability of moving from state A to state B.

4 **[Foundational]** • **Target: Diagonalization & Matrix Powers**

A transition matrix is $T = \left(\frac{2}{5}, \frac{3}{5}; \frac{3}{5}, \frac{2}{5}\right)$, rows/cols ordered A, B. Find $P(A|A)$, the probability of moving from state A to state A.

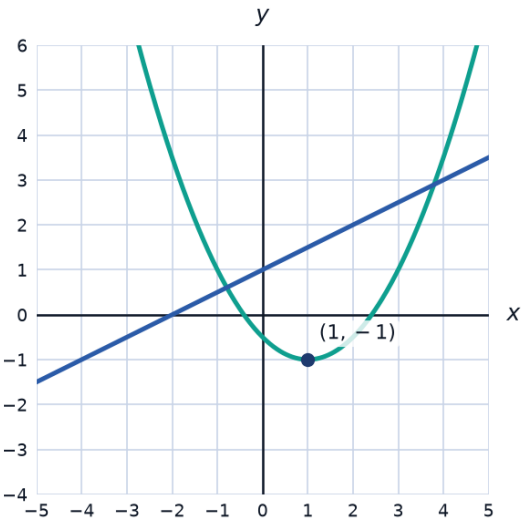
5 **[Foundational]** • **Target: Matrix Multiplication and Inverses**

A transition matrix is $T = \left(\frac{4}{5}, \frac{4}{5}; \frac{1}{5}, \frac{1}{5}\right)$, rows/cols ordered A, B. Find $P(A|B)$, the probability of moving from state B to state A.

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6 [Foundational] • Target: Determinants and Singularity

A transition matrix is $T = \left(\frac{2}{5}, \frac{4}{5}; \frac{3}{5}, \frac{1}{5}\right)$, rows/cols ordered A, B. Find $P(A|A)$, the probability of moving from state A to state A.



7 [Foundational] • Target: Eigenvalues and Eigenvectors

A transition matrix is $T = \left(\frac{3}{5}, \frac{1}{5}; \frac{2}{5}, \frac{4}{5}\right)$, rows/cols ordered A, B. Find $P(B|A)$, the probability of moving from state A to state B.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Diagonalization & Matrix Powers

A transition matrix is $T = \left(\frac{3}{5}, \frac{1}{5}; \frac{2}{5}, \frac{4}{5}\right)$, rows/cols ordered A, B. Find $P(B|B)$, the probability of moving from state B to state B.

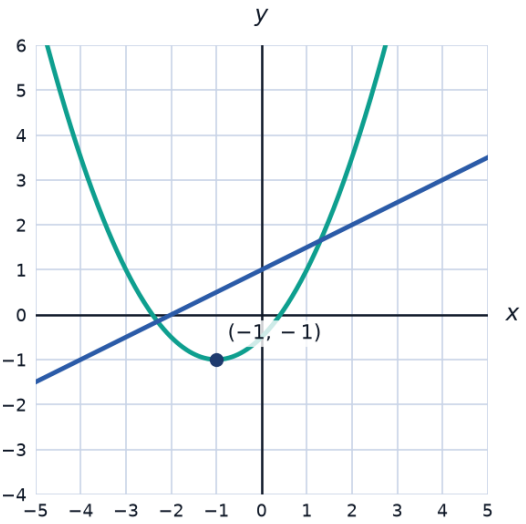
9 [Intermediate] • Target: Matrix Multiplication and Inverses

A transition matrix is $T = \left(\frac{1}{5}, \frac{2}{5}; \frac{4}{5}, \frac{3}{5}\right)$, rows/cols ordered A, B. Find $P(A|B)$, the probability of moving from state B to state A.

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10 [Intermediate] • Target: Determinants and Singularity

A transition matrix is $T = \left(\frac{4}{5}, \frac{1}{5}; \frac{1}{5}, \frac{4}{5}\right)$, rows/cols ordered A, B. Find $P(B|A)$, the probability of moving from state A to state B.



11 [Intermediate] • Target: Eigenvalues and Eigenvectors

A 2-state Markov chain has transition matrix $T = \left(\frac{9}{10}, \frac{1}{10}; \frac{3}{10}, \frac{7}{10}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

12 [Intermediate] • Target: Diagonalization & Matrix Powers

A 2-state Markov chain has transition matrix $T = \left(\frac{7}{10}, \frac{3}{10}; \frac{1}{10}, \frac{9}{10}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

13 [Intermediate] • Target: Matrix Multiplication and Inverses

A 2-state Markov chain has transition matrix $T = \left(\frac{7}{10}, \frac{3}{10}; \frac{3}{10}, \frac{7}{10}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

14 [Intermediate] • Target: Determinants and Singularity

A 2-state Markov chain has transition matrix $T = \left(\frac{9}{10}, \frac{1}{10}; \frac{1}{10}, \frac{9}{10}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

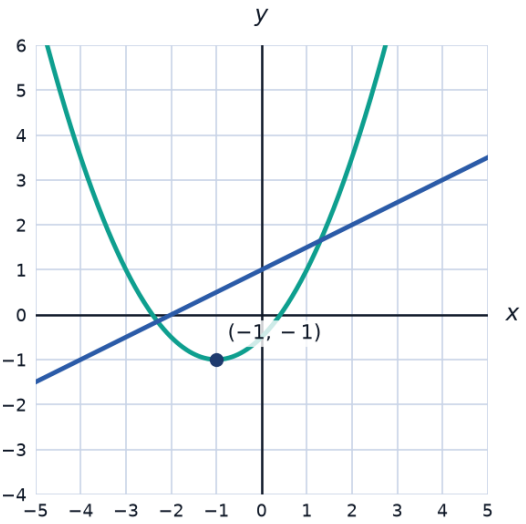
15 [Challenging] • Target: Eigenvalues and Eigenvectors • Cross-Lesson: 5.02 Differential Equations

A 2-state Markov chain has transition matrix $T = \left(\frac{3}{5}, \frac{2}{5}; \frac{1}{10}, \frac{9}{10}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

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16 [Challenging] • Target: Diagonalization & Matrix Powers • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

A system starts in state A. The transition matrix (rows/cols A, B) is $T = \left(\frac{4}{5}, \frac{1}{5}; \frac{3}{5}, \frac{2}{5}\right)$. Find $P(\text{state } A \text{ after 2 steps})$.



17 [Challenging] • Target: Matrix Multiplication and Inverses • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

A system starts in state A. The transition matrix (rows/cols A, B) is $T = \left(\frac{3}{5}, \frac{2}{5}; \frac{2}{5}, \frac{3}{5}\right)$. Find $P(\text{state } A \text{ after 2 steps})$.

18 [Challenging] • Target: Determinants and Singularity • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

A system starts in state A. The transition matrix (rows/cols A, B) is $T = \left(\frac{4}{5}, \frac{1}{5}; \frac{2}{5}, \frac{3}{5}\right)$. Find $P(\text{state } A \text{ after 2 steps})$.

19 [Challenging] • Target: Eigenvalues and Eigenvectors • Cross-Lesson: 6.03 Markov Chains (AI HL)

A system starts in state A. The transition matrix (rows/cols A, B) is $T = \left(\frac{3}{5}, \frac{2}{5}; \frac{4}{5}, \frac{1}{5}\right)$. Find $P(\text{state } A \text{ after 2 steps})$.

20 [Challenging] • Target: Diagonalization & Matrix Powers • Cross-Lesson: 1.01 Proof by Mathematical Induction

A system starts in state A. The transition matrix (rows/cols A, B) is $T = \left(\frac{2}{5}, \frac{3}{5}; \frac{4}{5}, \frac{1}{5}\right)$. Find $P(\text{state } A \text{ after 2 steps})$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Matrix Multiplication and Inverses

Compute the matrix product AB for $A = [[1, 2], [3, 4]]$ and $B = [[2, 0], [1, 3]]$.

a. $[[4, 6], [10, 12]]$

b. $[[2, 0], [3, 12]]$

c. $[[4, 10], [6, 12]]$

d. $[[3, 2], [4, 7]]$

22 [Foundational] • Target: Matrix Multiplication and Inverses

Find the inverse of the matrix $M = [[3, 5], [1, 2]]$.

a. $[[2, -5], [-1, 3]]$

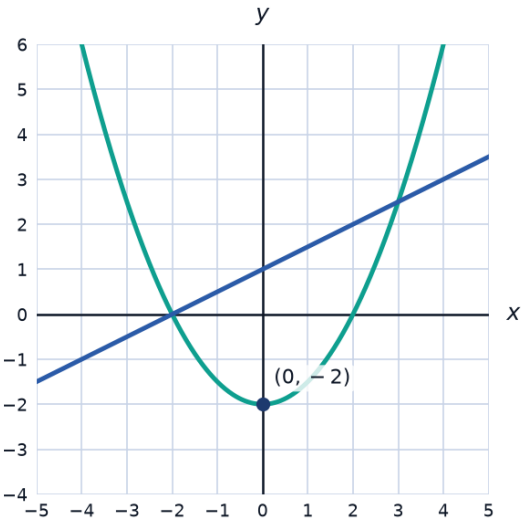
b. $[[-2, 5], [1, -3]]$

c. $[[3, -1], [-5, 2]]$

d. $[[1/3, 1/5], [1, 1/2]]$

23 [Foundational] • Target: Matrix Multiplication and Inverses

If A and B are invertible matrices of the same dimension, what is $(AB)^{-1}$?



a. $B^{-1}A^{-1}$

b. $A^{-1}B^{-1}$

c. BA^{-1}

d. AB^{-1}

24 [Foundational] • Target: Determinants and Singularity

For what value of k is the matrix $A = [[k, 6], [2, 3]]$ singular (non-invertible)?

a. $k = 4$

b. $k = 6$

c. $k = 0$

d. $k = 9$

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25 [Foundational] • Target: Determinants and Singularity

If A is a 3×3 matrix with $\det(A) = 5$, what is $\det(2A)$?

a. 40

b. 10

c. 20

d. 80

26 [Foundational] • Target: Determinants and Singularity

If $\det(A) = 4$, what is $\det(A^{-1})$?

a. $\frac{1}{4}$

b. -4

c. 4

d. 16

27 [Foundational] • Target: Eigenvalues and Eigenvectors

Find the eigenvalues of the matrix $A = [[4, 2], [1, 3]]$.

a. $\lambda = 5$ and $\lambda = 2$

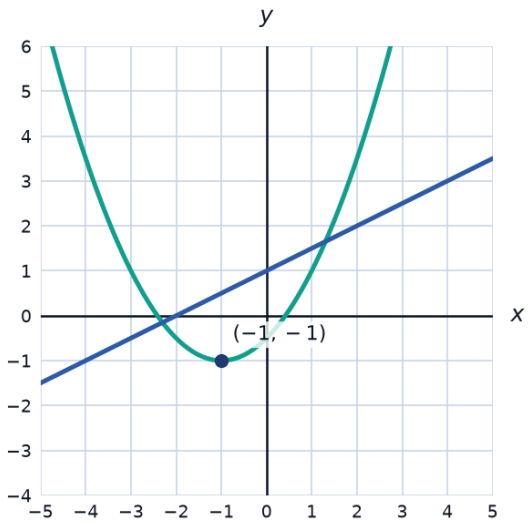
b. $\lambda = 4$ and $\lambda = 3$

c. $\lambda = 6$ and $\lambda = 1$

d. $\lambda = 7$ and $\lambda = 0$

28 [Foundational] • Target: Eigenvalues and Eigenvectors

Find an eigenvector corresponding to $\lambda = 5$ for $A = [[4, 2], [1, 3]]$.



a. $[2, 1]^T$

b. $[1, 2]^T$

c. $[1, -1]^T$

d. $[2, -1]^T$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Eigenvalues and Eigenvectors**

What is the sum of the eigenvalues of an $n \times n$ matrix A equal to?

a. The trace of A (sum of diagonal entries)

b. The determinant of A

c. The rank of A

d. n

30 **[Intermediate]** • **Target: Diagonalization & Matrix Powers**

If a matrix A is diagonalized as $A = PDP^{-1}$, how is the power A^n computed efficiently?

a. $A^n = PD^nP^{-1}$

b. $A^n = P^nD^n(P^{-1})^n$

c. $A^n = D^n$

d. $A^n = PDP^{-1}n$

31 **[Intermediate]** • **Target: Diagonalization & Matrix Powers**

If $D = [[2, 0], [0, 3]]$, what is D^4?

a. $[[16, 0], [0, 81]]$

b. $[[8, 0], [0, 12]]$

c. $[[24, 0], [0, 24]]$

d. $[[16, 81], [0, 0]]$

32 **[Intermediate]** • **Target: Diagonalization & Matrix Powers**

When is an $n \times n$ matrix A guaranteed to be diagonalizable?

a. When it has n linearly independent eigenvectors (or n distinct eigenvalues)

b. When its determinant is non-zero

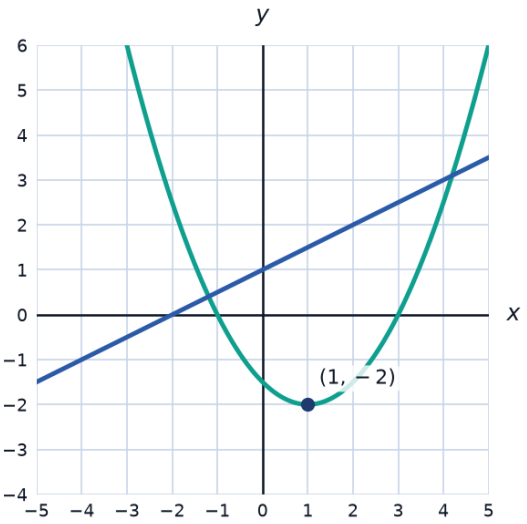
c. Only if it is a symmetric identity matrix

d. When all its entries are positive

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33 **[Intermediate]** • **Target: Matrix Multiplication and Inverses**

A 2-state Markov chain has transition matrix $T = \left(\frac{9}{10}, \frac{1}{10}; \frac{1}{5}, \frac{4}{5}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.



a. $\frac{2}{3}$

b. $\frac{2}{5}$

c. $\frac{9}{10}$

d. $\frac{11}{20}$

34 **[Intermediate]** • **Target: Determinants and Singularity**

A 2-state Markov chain has transition matrix $T = \left(\frac{9}{10}, \frac{1}{10}; \frac{2}{5}, \frac{3}{5}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

a. $\frac{3}{10}$

b. $\frac{1}{4}$

c. $\frac{1}{2}$

d. $\frac{4}{5}$

35 **[Intermediate]** • **Target: Eigenvalues and Eigenvectors**

A 2-state Markov chain has transition matrix $T = \left(\frac{3}{5}, \frac{2}{5}; \frac{1}{5}, \frac{4}{5}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

a. $\frac{1}{3}$

b. $\frac{4}{5}$

c. 1

d. $\frac{3}{10}$

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36 **[Intermediate]** • **Target: Diagonalization & Matrix Powers**

A 2-state Markov chain has transition matrix $T = \left(\frac{3}{5}, \frac{2}{5}; \frac{2}{5}, \frac{3}{5}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

a. $\frac{1}{2}$

b. $\frac{3}{10}$

c. 1

d. $\frac{2}{5}$

37 **[Intermediate]** • **Target: Matrix Multiplication and Inverses**

A 2-state Markov chain has transition matrix $T = \left(\frac{4}{5}, \frac{1}{5}; \frac{2}{5}, \frac{3}{5}\right)$ (rows/cols state 1, state 2). Find the steady-state probability of state 1.

a. $\frac{3}{5}$

b. $\frac{2}{5}$

c. $\frac{2}{3}$

d. 0

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Determinants and Singularity** • **Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)**

A system starts in state A. The transition matrix (rows/cols A, B) is $T = \left(\frac{3}{5}, \frac{2}{5}; \frac{3}{5}, \frac{2}{5}\right)$. Find $P(\text{state } A \text{ after 2 steps})$.

a. $\frac{9}{10}$

b. $\frac{3}{5}$

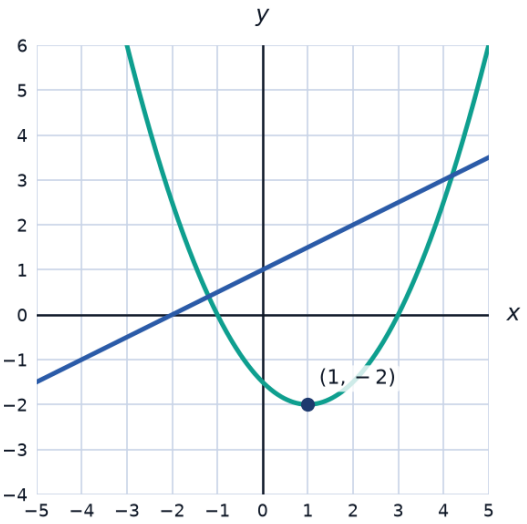
c. 1

d. $\frac{2}{5}$

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39 [Challenging] • Target: Eigenvalues and Eigenvectors • Cross-Lesson: 6.03 Markov Chains (AI HL)

A system starts in state A. The transition matrix (rows/cols A, B) is $T = \left(\frac{2}{5}, \frac{3}{5}; \frac{2}{5}, \frac{3}{5}\right)$. Find $P(\text{state } A \text{ after 2 steps})$.



a. $\frac{2}{5}$

b. $\frac{4}{5}$

c. $\frac{1}{5}$

d. 0

40 [Challenging] • Target: Diagonalization & Matrix Powers • Cross-Lesson: 1.01 Proof by Mathematical Induction

Eigenvalues of $(3, 1; 1, 3)$:

a. 2 and 4

b. 1 and 3

c. 3 and 3

d. 2 and 3

41 [Challenging] • Target: Matrix Multiplication and Inverses • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

A graph's adjacency matrix is $A = (0, 1, 1; 1, 0, 1; 1, 1, 0)$ (vertices 1,2,3). Find the number of walks of length 2 from vertex 1 to vertex 1 (the $(1, 1)$ entry of A^2).

a. 1

b. 0

c. 2

d. 4

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42 [Challenging] • Target: Determinants and Singularity • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

A graph's adjacency matrix is $A = (0, 1, 1; 1, 0, 1; 1, 1, 0)$ (vertices 1,2,3). Find the number of walks of length 2 from vertex 2 to vertex 2 (the $(2, 2)$ entry of A^2).

a. 2

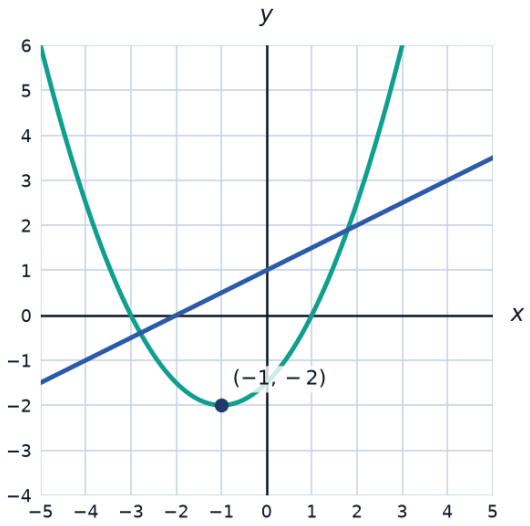
b. 3

c. 1

d. 5

43 [Challenging] • Target: Eigenvalues and Eigenvectors • Cross-Lesson: 1.04 Partial Fractions

A graph's adjacency matrix is $A = (0, 1, 1; 1, 0, 1; 1, 1, 0)$ (vertices 1,2,3). Find the number of walks of length 2 from vertex 2 to vertex 3 (the $(2, 3)$ entry of A^2).



a. 3

b. 1

c. 4

d. 2

44 [Challenging] • Target: Diagonalization & Matrix Powers • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

A graph's adjacency matrix is $A = (0, 1, 1; 1, 0, 1; 1, 1, 0)$ (vertices 1,2,3). Find the number of walks of length 3 from vertex 2 to vertex 3 (the $(2, 3)$ entry of A^3).

a. 6

b. 2

c. 0

d. 3

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45 [Challenging] • Target: Matrix Multiplication and Inverses • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

A graph's adjacency matrix is $A = (0, 1, 1; 1, 0, 1; 1, 1, 0)$ (vertices 1,2,3). Find the number of walks of length 3 from vertex 1 to vertex 1 (the $(1, 1)$ entry of A^3).

a. 2

b. 3

c. 0

d. 4

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